data is quantitative information of a method deals with that quantitative information.

1.2 MEANING OF STATISTICS

Statistics has been defined differently by different users. For a common man, statistics means data, facts or measurements. Some people define it as study of figures. Diagrammatic or Graphic Representation of facts is also understood as statistics. Also some consider it as analysis of figures for forecasting or drawing inferences.

However, it is true that statistics is the aggregate of facts expressed numerically, but it is an inadequate description of it as discipline of study.

The word statistics is used in two different senses :

(1) Plural Sense (Statistics as Data) (2) Singular Sense (Statistics as a Science)

Statistics is used for collecting, organizing, presenting, analyzing and interpreting numerical data or data that may be converted into numerical form. It involves variety of methods to assist in the analysis of data for the purpose of decision-making. The important area of the application of these statistical techniques is in business and commerce, which is called study of business statistics.

This book deals with business statistics, so here we discuss how statistical data, tools and methods are used for taking decision in the business to run it efficiently.

1.3 DEFINITIONS OF STATISTICS

There are as many definitions of statistics as there are authors. In 1869 **Quetelet** collected 180 definitions of statistics. *The definitions of statistics can be broadly classified into two categories :*

(I) Definitions in the Plural Sense (as Statistical Data)

(II) Definition of Singular Sense (as Statistical Methods)

In first category, definitions lay emphasis on statistics as data while in second category, it emphasize on statistics as a scientific method.

I. Statistics as Data (in Plural Sense)

Statistics in the plural sense implies as set of numerical facts or figures, systematically collected for some definite purpose under investigation. It refers to collection of numerical data or figures obtained from any field of enquiry such as statistics of prices, food, production, expenditure, income, population, educational institution etc.

Data in numerical form is referred as statistical data, but it should be noted that any set of numerical data cannot be regarded as statistics. It is rightly said that, *"All statistics are numerical data, but all numerical data are not statistics."*

These are certain characteristics which must be satisfied by a given set of numerical data to termed it as statistics. For the purpose of understanding characteristics, we first go through same definitions of statistics in the plural sense :

1. "Statistics are numerical statements of facts in any department of enquiry placed in relation to each other." **–Dr. A.L. Bowley**
2. "Statistics are classified facts representing the conditions of the people in a state ..... especially those facts which can be stated in numbers or in tables of numbers or in any tabular or classified arrangements." **–Webster**
3. "By statistics we mean quantitative data affected to a marked extent by multiplicity of causes." **–Yule and Kendall**
4. "Measurements, enumerations or estimates of natural or social phenomena systematically arranged so as to exhibit their inter-relations." **–L.R. Conner**
5. "Statistics are numerical descriptions of quantitative aspects of things and they take the form of counts or measurements." **–Wallis and Roberts**

All the above definitions cover only some but not all characteristics of statistical data.

By **Prof. Horace Secrist,** *"Statistics may be defined as the aggregate of facts affected to a marked extent by multiplicity of causes, numerically expressed, enumerated or estimated according to a reasonable standard of accuracy, collected in a systematic manner, for a predetermined purpose and placed in relation to each other."*

This definition of **Secrist** can be taken as comprehensive and adequate definition of statistics, since it covers most of the characteristics of statistical data.

1.3.1 Characteristics of Statistics as Numerical Data

Some important characteristics of good statistical data are :

1. **Statistics are aggregates of facts :** Single or unrelated facts do not throw light on any problem. So a single figure related to age, height, weight, birth, death, sale, purchase etc. even though numerically stated cannot be called statistics. But a group of figures related to ages, heights, weights, births, deaths, sales, purchases etc. would be called *Statistics*. For example, a single fact say 500 lakh metric tons production of copper in India during the year 2000 is of no interest to a statistician, as the figure is unrelated and incomparable, but if production figures for several consecutive years are available, then they could be known as *statistics* as the figures are aggregate and comparable.
2. **Statistics are affected to a marked extent by multiplicity of causes :** It means that collection of facts and figures are affected to a considerable extent by a number of forces operating together. It is usually not possible to study the effects of anyone of these forces separately. For example, statistics of production of rice is affected by the rainfall, fertility of soil, quality of seeds and manure, method of cultivation etc. It is very difficult to study separately, the affect of each of these forces on the production of rice. Also, it is true that price level in market is influenced by many factors like demand, supply, import, export, capital etc. and it is difficult to study separately the effects of each of these facts on price level.
3. **Statistics should be numerically expressed :** There are generally two types of statements of facts, quantitative and qualitative. Quantitative statements expressed in numbers but qualitative statements such as 'India is a poor country', 'prices are rising', 'production of wheat is not sufficient', do not express statistics. The reason is that such statements are vague and constitute nothing. So only quantitative statements expressed numerically are statistics.
4. **Statements should be enumerated or estimated according to reasonable standard of accuracy :** Statistical data are collected either by enumeration or estimation. Enumeration of data is actual counting of data. Estimation cannot be as precise and accurate as actual counts or measurements. Enumeration is done only when the field of enquiry and units to be studied are limited e.g., to find the number of students in a college, data is collected by actual country or to find out the weight of students of a particular class, then data is collected by actual measurement. If the scope of enquiry is very wide or the units are very large or the cost of measurements is high, then enumeration is usually avoidable or not possible and in such cases, data can only be estimated by examining only a fraction of the total area of enquiry.

   When statistics are to be obtained by measurement of units, it is necessary to maintain a reasonable degree or standard of accuracy, in measurements. The standard of accuracy depends upon the purpose of enquiry. It cannot be uniform for all types of enquiries. For example, for weighing rice, a few grams add or increase may not make much difference, but for weighing gold, it matters a lot.

According to **Dr. A.L. Bowley,** *"Great numbers are not counted correctly to a unit, they are estimated."* It is normal that the estimated figures cannot be absolutely accurate. Some errors or differences in counting and measurement are likely to take place, but it should be as minimum as possible. The reasonable standard of accuracy will depend upon the nature and purpose of enquiry. **Prof. Secrist** has stated that, *"What are reasonable standards depend upon the purpose which the statistics are to serve."*

5. **Statistics should be collected in a systematic manner :** The statistical data should be collected in a systematic manner to reach at precise and proper conclusion. The data collected in haphazard manner may lead to misleading and fallacious conclusions. According to **Prof. Secrist,** *"Stray and loose bits of quantitative information, hear say and unrelated material gleaned here and there, from indiscriminate sources having no common basis of selection, even when numerical can be termed statistics only by confusion of terms."* A suitable plan of data collection should be prepared beforehand and trained personnel should be used to roll the required data. Also care should be taken to reduce the personal bias and prejudices to minimum, so that reasonable level of accuracy be attained in statistics collection.
6. **Statistics should be collected for pre-determined purpose :** Statistics should be collected for a specific, well defined and pre-determined purpose. A plan of enquiry should be prepared according to such purpose, otherwise it might happen that the unnecessary statistics are collected while the necessary statistics are left out. It would only be wastage of time, money and energy as well as wrong and fallacious conclusions are drawn for undertaken enquiry. For example, if we are to collect statistics of agricultural production, we must decide beforehand, the regions, commodities and periods for which the data is required.
7. **Statistics should be placed in relation to each other :** The statistical data that are collected should be comparable with each other. Comparison is one of the function of statistics. For this purpose, the data must be homogeneous and not heterogeneous. This can be achieved by collecting data in identical manner for different periods or for different places or for different conditions. For example, population of India at a particular point of time can be compared with the population of other countries like China, U.K., U.S.A. etc. (is case of homogeneous data). Heterogeneous data like marks of students of a particular class and height of trees, can never be placed in relation to each other and thus cannot be compared for analysis.

In the absence of the above characteristics, numerical data cannot be called statistics and hence we can say *"all statistics are numerical data but all numerical data are not statistics."*

II. Statistical Methods (in Singular Sense)

Numbers play an essential role in Statistics. They provide data as raw material, which is to be processed through the use of appropriate statistical methods. So, statistics in the singular sense is a science, which provide statistical methods that can be used to organize, present, analyse and interpret the information effectively.

Therefore we refer the term *"Statistics"* as the subject of statistical activity dealing with the theories and methods of collection, analysis and interpretation of data. It is applied for handling numerical data in all branches of business, commerce, trade, science or human activities where measurement or counting is possible. These methods finally help in taking decisions on the face of uncertainty and testing hypothesis.

To know about the various aspects of statistics, some of the *important definitions of the science of statistics are given below :*

1. "Statistics may be called the science of counting." **–A.L.Bowley**
2. "Statistics may rightly be called the science of averages." **–A.L. Bowley**
3. "Statistics is the science of the measurement of social organism, regarded as a whole in all its mainfestations." **–A.L. Bowley**
4. "Statistics is the science of estimates and probabilities." **–Prof. Boddington**
5. "Statistics may be defined as the collection, presentation, analysis and interpretation of numerical data." **–Croxton and Cowden**
6. "Statistics is the science which deals with the methods of collecting, classifying, presenting, comparing and interpreting numerical data collected to throw some light on any sphere of enquiry." **–Seligman**

Most of the above definitions are inadequate definitions, but **Croxton and Cowden** has given a very simple and concise definition of statistics.

This definition clearly points out four stages in a statistical investigation namely :

(i) Collection of data.
(ii) Presentation of data.
(iii) Analysis of data.
(iv) Interpretation of data.

However one more stage may be added to above stages as **'Organization of data.'**

Thus statistics may be defined as the science of collection organization, presentation, analysis and interpretation of numerical data.

1.3.2 Stages in Statistical Investigation

According to above definition there are five stages in statistical investigation.

1. **Collection of Data :** Under this stage, the relevant data is collected from various sources. But utmost care should be taken while collecting data. Data should be taken in a planned manner. Without proper planning, the collected data may not be suitable for the purpose of enquiry and lot of time and money may be wasted. The collected data should be precise, exact and meaningful. The data may be available from primary or secondary sources. Data is said to be primary when it is collected originally by an investigator or agency and it is said to be secondary when it is collected from published or unpublished sources, which have already collected data.
2. **Organisation of data :** Data collected by investigator is either primary or secondary. Generally data collected from published sources (i.e. secondary) is in organised form, but still investigator may have to reorganize it according to his needs. However a huge mass of figures that are collected from a survey i.e. primary data, need compulsory organisation.

The organisation of data is done in three parts :

(a) *Editing* : Editing is done for detecting error, inaccuracies, irregularities, inconsistencies in the data collected

(b) *Classification* : The purpose of classification is to arrange the data into same groups or classes according to some common characteristics possessed by the items constituting the data. Unclassified data is of no use. **J.R. Hicks** has rightly said, *"Classified and arrange facts speak themselves, unarranged – they are as dead as mutton."*

(c) *Tabulation* : The last step in the organization of data is tabulation. The purpose of tabulation is to arrange the data in rows and columns so that there is absolute clarity in the data presented.

3. **Presentation of Data :** After the data have been collected and organized, it is ready for presentation. Data presented in systematically and precise manner make statistical analysis easily understandable. The organised data can be presented with the help of statistical tables, diagrams, graphs etc. It includes tables, bar graphs, pie-charts, flow charts etc.

4. **Analysis of data :** After collection, organisation and presentation, the next steps is to analyse data. The main objective of analysis is to prepare data in such a fashion which helps the investigator to dig out useful information to arrive at certain definite conclusions. The methods most commonly used are *Measures of Central Tendency, Measures of Dispersion, Skewness, Correlation, Regression, Interpolation etc.*

5. **Interpretation of Data :** Interpretation or drawing valid conclusions on the basis of data analysed is the last stage in a statistical investigation. It requires high degree of skill, experience and common sense. If the analysed data is not interpreted carefully, it may lead to wrong and fallacious conclusions, which further defeat the basic objective of enquiry.

1.4 FUNCTIONS OF STATISTICS

Statistics is used to present the numerical data in such a form that is easily understandable by human mind and used easily to make comparisons, derived valid conclusions etc. **Robert W. Buges** describes the functions of statistics in these words, *"The fundamental gospel of statistics is to push back the domain of ignorance, prejudice, rule of thumb, arbitrary or premature decisions, traditions and dogmatism and to increase the domain in which decisions are made and principles are formulated on the basis of analysed quantitative facts."*

The main functions of statistics are as follows :

1. **To Present Facts in Numerical Figures :** One of the most important function of statistics is to present the facts relating to given phenomena in numerical figures. Numerical presentation helps in having a better understanding of the nature of problem. Facts expressed in words are often vague and are likely to be understood differently by different users.

2. **To Present Complex Facts in a Simplified Form :** The raw data is often unwieldy and complex. The function of statistics is to simplify a large mass of complex data and present it in a more understandable and intelligible form. **Dr.**

Bowley states that, *"The object of a statistical estimate of complex groups is to present an outline to enable the mind to comprehend with a single effort the significance of the whole."* In the words of **W.I. King**, *"It is for the purpose of simplifying these unwieldy masses of facts that statistical science is useful. It reduces them to numerical totals or averages which may be abstractly handled like any other mere numbers. It draws pictures and diagrams to illustrate general tendencies and thus in many ways adapt these groups of ideas to the capacity of our intellects."*

The simplification is achieved by the summarization of data so that broad-features of the given problem are brought into our focus. Statistical technique like average, dispersion, graph, diagram etc. make the numerical data more understandable.

3. **To Enlarge Individual Knowledge and Expertise :** Statistical methods provide us such a master key that we can use these methods anywhere and can study any problem in its correct context and prospective. Statistics enlarges human knowledge and expertise in the solution of various problems. In the words of **A.L. Bowley,** *"The proper function of statistics, indeed is to enlarge individual experience."* In the words of **Whipple,** *"Statistics enables one to enlarge his horizon."* Many fields of knowledge would have remained closed to mankind foreever without the efficient and useful techniques and methodology of the science of statistics. The concepts such as national income, population, agricultural production, industrial production, poverty, unemployment etc. can be understood clearly and precisely using statistical techniques.
4. **To Provide Comparative Study :** The science of statistics does not merely mean counting but also comparison. Comparison is easy and in meaningful quantitative terms. Simplified data gives us a helping hand in making comparative study of two or more phenomena. No conclusions can be drawn without comparisons. It is said that everything is understood in a relative sense. **Dr. A.L. Bowley** rightly states, *"A chief practical use of statistics is to show relative importance, the very thing which an individual is likely to misjudge. Statistics are almost always comparative."* Statistical techniques of averages, variances, coefficients, ratios, percentages, trends, graphs and diagrams offer the various means of comparison.
5. **To Study Relationship Between Different Facts :** Statistics helps to investigate whether the two or more phenomena are correlated. It also find out the degree of the relationship and nature of relationship (i.e. positive or negative). For example the relationship between demand and supply, income and expenditure, rainfall and agricultural production, age of husbands and their wives etc. can be studied with the help of correlation anaysis. Also, techniques of regression analysis helps to find mathematical relation fitted in the given data.
6. **To Formulate Policies in Different Fields :** Statistics helps in formulating policies in various fields, specially in social, economic and business fields. Various laws have been developed on the basis of statistics. **e.g. Malthus,** *"Theory of Population"* and **Engel's** *"Law of Family Expenditure"* are due to the statistics collected. Policies in every field, whether private or government, are formulated on the basis of statistical data. For example government policies regarding taxation, prices, investments, imports and exports, social security, etc are guided by

statistics and also a firm can know the tastes and preferences of the consumers and decide to make its product accordingly. **Vital statistics** comprising birth and mortality rates help in assessing future growth in population and this information is very useful in designing any scheme of family planning.

7. **To Help in Forecasting :** The success of planning and policies formulation by the government or a business depends largely on its accuracy in future forecasts. Statistics provides scientific techniques for extrapolation and forecasting. Knowledge of future trends is helpful in framing plans. For example, forecast can be made with regard to changes in future prices, food production, supply of manpower, population growth etc. results in **five year plans** in India. Various techniques used for forecasting are time series analysis, regression analysis, extrapolation theory etc.
8. **To Measure the Effects :** Statistics helps not only in formulation of policy but also in measuring the effects of the policy implementation. The effects of a change in the rate of interest in banks or G.S.T., Sales Tax etc. can be thoroughly studied with the help of statistics.
9. **To Classify Data :** Statistics helps in classification of data into some groups or classes according to some common characteristics between them. It is necessary for analyzing and interpretation of data for drawing conclusions easily with respect to the phenomena under study.
10. **To Formulate and Test Hypothesis :** Statistical methods are extremely helpful in formulating and testing hypothesis and thereby developing new theories. A hypothesis is a statement about same characteristic of population (or universe). For example, hypothesis like the particular dice is fair or not, whether average height of students of a particular college is 5.8' feets or not, etc. It is possible to test the validity of above statements by using statistical techniques.
11. **To Provide Techniques for Making Decisions Under Uncertainty :** Many times, we face a situation of uncertainty or we have to decide one alternative from many ones.

For example, a businessman might face the situation of uncertainty in choosing an investment plan, to start production of new product, to decide the price for his new product etc. The answer to such problems are provided by the statistical techniques of decision making under uncertainty.

1.5 USES AND IMPORTANCE OF STATISTICS

SCOPE

In the words of **H.G. Wells,** *"Statistical thinking will one day be as necessary for efficient citizenship as the ability to read and write."* These words are proving true in present day life. Today it is perhaps difficult to imagine a field of knowledge which cannot use statistics. **M.J. Moroney,** writes, *"Whoever you are, if your work calls for interpretation of data, you may be able to do without statistics, but you doesn't do so well."*

It is impossible to understand Psychology, Sociology, Economics, Finance and Physical Sciences even at their basic stage, without understanding the meaning of various statistical techniques like average, dispersion, correlation etc. Knowingly or unknowingly, every person employ statistical methods in taking many decisions. For example, if a person wish to purchase a laptop, then he studies the pricelists of various companies for making a choice. He uses the idea of range (measure of dispersion) within which the prices vary. The

whole business of insurance rests on statistical facts. That is why **Tippet** remarked, *"Statistics affects everybody and touches life at many points."* It is due to statistics, that many fields have progressed in the past and are still progressing. Statistics has embraces all sciences and diversified fields such as agriculture, industry, psychology, sociology, biometry, geology, planning, economics, business, commerce, management, insurance, accountancy, auditing etc.

We shall discuss briefly the importance of statistics in the following major area :

1. Importance of Statistics to the State and Administration :

We know that the subject of statistics originating for helping the ancient rulers to estimate their military and economic strength. But gradually its scope was enlarged to solve other problems relating to political activities of the state. Now in modern era, the states have grown and the concept of welfare state has taken roots to undertake the responsibilities of social and economic welfare of its people. For this a lot of statistical data is required to know about the social and economic conditions of its people. These statistics help in framing suitable policies and its implementations. Statistics helps the government during both peace and war time. It is observed that the government is the biggest collector and user of statistical data in most of the countries. All ministers and departments of the government like Finance, Defence, Transport, Railways, Food, Post and Telegram, Commerce, Agriculture, Small Scale and Large Scale Industries etc. depend heavily on statistical data for their efficient functioning. For example, Railway department cannot solve the problem of rail transport in Northern India unless it knows how many trains are operating at present, how many are required in total and how much trains be added additionally to selected routes. The most important function of the state is to prepare annual general budget which can be formulated only on the basis of statistics. The reports of government committees and commission appointed for specific purpose are also based on statistics. Moreover, in a democratic form of government, various political parties are also guided by the statistical analysis regarding their popularity in the people. Thus it can be stated that statistics is vital for the state.

2. Importance of Statistics to the Business, Industry and Commerce

With the gradual increase in Industrialization and consequent expansion of trade, businessmen can no longer depend on the old system of working and leave their future on chance. Business is regarded as a profession of forecasting which requires high degree of skill to recognize the pulse of future. Success in business depends upon the precision in forecasting. A businessman or industrialist must make a proper and scientific plans, prepare themselves for competitive markets, properly analysis the past record in order to forecast the future business conditions. The traders and manufacturers have to depend on a variety of factors e.g., past sales records, present labour conditions and prices of raw materials and similar products available in the market etc. All these factors must be taken in account before fixing the price of new product.

Every manufacturer must know in advance 'how much is to be produced', 'how much raw material and workers are needed to produce the estimated quantity' and 'what should be the quality, type, size, colour, grade etc. of the product to be manufactured.' etc. He has to employ statistical techniques in estimating trend of prices, trend of economic activities etc. **Buddington** says, *"In order to succeed in any business today, the businessman must study all the factors which enter into production, buying and selling, exporting and importing of goods in which he deals."* He further says, *"A successful businessman is one who*

estimates most closely approach accuracy." Accuracy can be achieved only by proper analysis of past records with the help of statistical methods. Even the manufacturers, producers, industrialists take decision about the quantity of future production on the basis of past records and trends.

With the growing size and ever-increasing cut throat competition, the problems of the business, industry and trade sectors are becoming complex and they are increasing the use of statistics in decision making. Statistical methods are applied in market and product research, investment policies, quality control of manufactured products, economic forecasting etc. Statistics are equally important to the stock and share brokers, investors etc. Statistical methods of analysis are helpful in the marketing function of an enterprise. It helps in market research, advertisement campaigns in comparing sales performance.

Some of the statistical techniques applied in business, commerce and industry are :

(i) To find main characteristics of available records, the various statistical measures of average, dispersion, skewness, correlation etc. are applied. This calculations facilitates the comparison between different time periods or geographical areas.

(ii) **'Time series'** analysis helps in isolating the components of variation noticed in chronological records of prices, sales, profits, production etc. Various trends like seasonal, cyclic helps in future forecasts.

(iii) **Index numbers** of price, production, cost of living, business activities are known as 'business barometers.'

(iv) **Regression analysis** helps in establishing a relationship between various economic factors e.g., demand and per capita income, production and sales etc.

(v) **Statistical quality control** is now being used in industry for establishing quality standards for products and in the maintenance of quality during production.

We may conclude that industry and business involve risks and it is better to calculate risk, so that precautions can be taken before hard. With the help of statistical tools, these risks can be calculated with high degree of accuracy.

3. Importance of Statistics in Economics

The interaction between statistics and economics was first observed by **William Petty** in his book, **'Political Arithmetic'** in end of 17th century. But effective use of statistics in formation of economic theories and policies took a long time. Theories were based on deductive logic only and also the statistical techniques were not as much advanced to be used in other disciplines.

In year 1890, **Prof. Alfred Marshall** observed that, *"Statistics are the straws out of which I, like every other economist, have to make bricks."* This proves the significance of statistics in economics. Economic problem always involve facts that are capable of being expressed numerically e.g., prices of commodities, saving and investment of income, bank deposits, manufacturing output, volume of trade, agricultural wages etc. Statistical data and statistical methods are of immense help in the understanding economic problems and formulating the economic policies. Without statistical data, no economic problem can be studied. Economists collect and analyse statistical data about economic problems and apply statistical techniques to formulate the policies and arrived at certain decisions. Most of the laws and principles in economics like *Laws of Demand and Supply, Consumer Surplus, Law of Diminishing Marginal Utility, Law of Diminishing Marginal Returns, the Theory of Related Preference etc.* can only be expressed and understood with appropriate statistical data.

Economics is concerned with production, distribution, consumption, exchange etc. Statistics of production help us to adjust the supply according to demand. Such statistics are the measures of productivity of a country. With the help of statistics of production, progress of a nation can be measured year after year. Census of production has become an important feature of every nation. In the distribution too, statistics play a vital role. Statistics is a tool in the hands of economists with which they compute national income and wealth and its distribution among people of the country. Statistics of consumption enable us to find out the way in which the people of different classes of society spend their incomes. It also helps to know their standard of living and taxable capacity. In the field of exchange, statistics is very useful in studying markets, laws of prices based on demand and supply, cost of production etc.

A statistical approach to an economic problem not only leads to its correct description but also indicates how it is to be tackled. The increasing importance of statistics in the study of economic problems has resulted in a new branch of study called **Econometrics.** In the words of **Tippet,** *"A day might came when the departments of economics in the universities will go out of the control of economic theoreticians and came under the control of statistical workshops, in the same manner as the department of physics and chemistry have came under the control of experimental laboratories."*

4. Importance of Statistics in Planning

"Economic planning means making of major economic decisions – what and how much is to be produced and to whom it is to be allocated – by the conscious decisions of a determinate authority on the basis of a comprehensive survey of the economy as a whole." It implies a conscious and efficient action to attain pre-determined economic aim. Today, is an age of planning, statistics plays a very vital role in planning. Statistics is indispensable in planning .

In the words of **Tippet,** *"Planning is the order of the day and without statistics, planning is inconceivable."* Planning is indispensable for achieving faster rate of growth through the best use of national resources. Using statistical techniques, it is possible to access the amount of various resources available in the economy and accordingly planning, would be done. Statistical techniques and principles can also guide the Government in adopting suitable policies for betterment of economic problems.

The importance of statistics in planning can be explained as follows :

(i) Statistics is essential to draw plans for economic development to achieve the objective of utilizing the economic resources in the best possible ways.

(ii) To know the position of economic development of our country as compared to other countries, statistics are of great help. The problems like unemployment, over-population, lack of industrialization, low rate of capital formation are characteristics of underdeveloped and underdeveloped countries.

(iii) Use of statistical technique in the evaluation of executed plans is necessary. The difference between the largest and achieved result should be worked out.

(iv) Statistics are used to determine priorities in different sectors for development. This can be done by examining the structure and importance of consumption, production, exchange and distribution sectors of the economy.

Economic planning is done to achieve pre-determined targets. They have to be expressed in numerical facts. Financial planning is a part of economic planning. Financial

resources are moved through taxation, public saving, deficit, financing etc. Planning and budgetary are integrated. All these functions can be preformed efficiently with the help of statistics.

For planning, sometimes said that, "Planning without statistics is a ship without rudder and compass."

5. Importance of Statistics in Banks and Insurance Companies

Banks use statistical techniques to decide the amount for reserves that is to be kept with them in cash to meet the requirements of day to day transactions and demand of their customers for money is satisfied. Otherwise public confidence in the banks is lost. Further, various policies of investment and sanction of loans are also based on the analysis provided by statistics.

The business of insurance is based on the studies of life expectancy in various age groups. Insurance companies have to prepare life tables and calculate premium rates. All this work involves the use of statistical principles and methods.

6. Importance of Statistics in Research

Statistics is indispensable in research work. According to remark of **Wallis and Roberts,** *"Statistics is not a body of substantive knowledge but a body of methods for obtaining knowledge."* Most of the advancement in knowledge has taken place due to research conducted with the help of statistics. So statistics is known as the body of methods for obtaining knowledge. The statistical methods simplify the mass of complex data, present facts, facilitate comparisons and help in testing hypothesis, frame policies and make predictions.

Introduction to Data

Data Collection enables a team to formulate and test working assumptions about a process and develop information that will lead to the improvement of the key quality characteristics of the product or service. Data Collection improves your decision-making by helping you focus on objective information about what is happening in the process, rather than subjective opinions. The purpose of data collection is to obtain information to keep on record, to make decisions about important issues, to pass information on to others. Primarily, data are collected to provide information regarding a specific topic.

The collection and analysis of secondary data help to define the marketing research problem and develop an approach. In addition, before collecting primary data, the researcher should locate and analyze relevant secondary data. Thus, secondary data can be an essential component of a successful research design. Secondary data can help in sample designs and in the details of primary research methods. In some projects, research may be largely confined to the analysis of secondary data because some routine problems may be addressed based only on secondary data. In addition, given the huge explosion of secondary data sources available, sufficient data may be accessed to solve a particular marketing research problem.

Prior to any data collection, pre-collection activity is one of the most crucial steps in the process. It is often discovered too late that the value of their interview information is discounted as a consequence of poor sampling of both questions and informants and poor elicitation techniques. After pre-collection activity is fully completed, data collection in the field, whether by interviewing or other methods, can be carried out in a structured, systematic and scientific way. A formal data collection process is necessary as it ensures that data gathered are both defined and accurate and that subsequent decisions based on arguments embodied in the findings are valid.

Meaning of Data

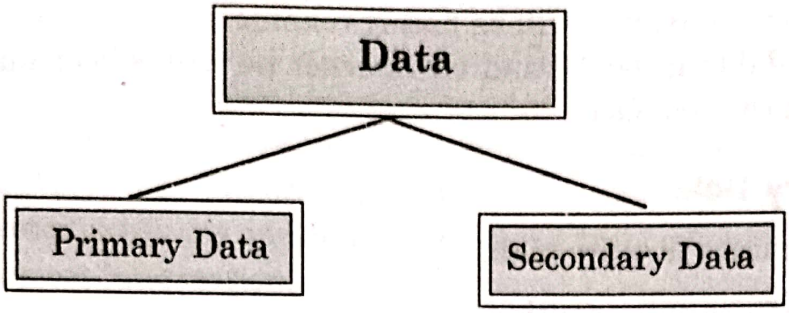
Data is the facts in raw or unorganized form such as alphabets, numbers or symbols that refer to or represent conditions, ideas or objects. This represents facts and statistics which are collected together for reference or analysis.

Collection of Data

Data collection is the process of collecting of data from surveys or from independent or networked locations via data capture, data entry or data logging. This is a term used to describe a process of preparing and collecting data, as part of a process improvement or similar project. The data may be collected for the whole population or for a sample drawn from the population is called collection of data.

Business Data Sources

The data can be classified into two categories:



Primary Data

Primary data is data that has not been previously published, i.e., the data is derived from a new or original research study and collected at the source, e.g., in marketing, it is information that is obtained directly from first-hand sources by means of surveys, observation or experimentation.

Importance of Primary Data

Importance of Primary data cannot be neglected. A research can be conducted without secondary data but a research based on only secondary data is least reliable and may have biases because secondary data has already been manipulated by human beings. In statistical surveys it is necessary to get information from primary sources and work on primary data: for example, the statistical records of female population in a country cannot be based on newspaper, magazine and other printed sources. One such source is old and secondly they contain limited information as well as they can be misleading and biased.

(i) ***Validity:*** Validity is one of the major concerns in a research. Validity is the quality of a research that makes it trustworthy and scientific. Validity is the use of scientific methods in research to make it logical and acceptable. Using primary data in research can improves the validity of research. First hand information obtained from a sample that is representative of the target population will yield data that will be valid for the entire target population.

(ii) ***Authenticity:*** Authenticity is the genuineness of the research. Authenticity can be at stake if the researcher invests personal biases or uses misleading information into the research. Primary research tools and data can become more authentic if the methods chosen to analyze and interpret data are valid and reasonably suitable for the data type. Primary sources are more authentic because the facts have not been overdone. Primary source can be less authentic if the source hides information or alters facts due to some personal reasons. There are methods that can be employed to ensure factual yielding of data from the source.

(iii) ***Reliability:*** Reliability is the certainty that the research is enough true to be trusted on. For example, if a research study concludes that junk food consumption does not increase the risk of cancer and heart diseases. This conclusion should

have to be drawn from a sample whose size, sampling technique and variability is not questionable. Reliability improves with using primary data. In the similar research mentioned above if the researcher uses experimental method and questionnaires the results will be highly reliable. On the other hand, if he relies on the data available in books and on internet he will collect information that does not represent the real facts.

Merits of Primary Data

i) Degree of accuracy is quite high.

ii) It does not require extra caution.

iii) It depicts the data in great detail.

iv) Primary source of data collection frequently includes definitions of various terms and units used.

v) For some investigations, secondary data are not available.

Demerits of Primary Data

i) Collection of data requires a lot of time.

ii) It requires lot of finance.

iii) In some enquiries it is not possible to collect primary data.

iv) It requires a lot of labor.

v) It requires a lot of skill.

Methods / Sources of Primary Data

Primary data are collected by the following sources:

1. Personal Investigation/enquiry or Direct Personal observation

The information is collected by direct personal interviews. The data collected in this way is usually accurate and reliable. It is necessary that investigator has to keep observing while collecting data. The information thus obtained is called first-hand or original in charter. So this method is useful when the scope of enquiry is small.

Merits

(i) Response is more encouraging and more people are willing to supply information when approached personally.

(ii) Accurate and additional information may be collected.

Demerits

(i) It is expensive and time-consuming.

(ii) There are chance of personal prejudice and bias.

2. Indirect Oral Interviews

This is an indirect method of collecting primary data. Here information are not collected directly from the source but by interviewing persons closely related with the problem. This method is applied to apprehend culprits in case of theft, murder etc. The information relating to one's personal life or which the informant hesitates to reveal are better collected by this method. Here the investigator prepares a small list of questions relating to the enquiry. The answers (information) are collected by interviewing persons well connected with the incident. The investigator should cross-examine the informants to get correct information.

Merits

(i) This method is time saving and involves relatively less cost.

(ii) Accurate and additional information may be collected.

(iii) Vast area can be covered.

Demerits

(i) The third party may not be willing to co-operate with the investigation.

(ii) The accuracy of the information largely depends upon the integrity of the investigator.

3. Mailed Questionnaire Method

A questionnaire is a document prepared by the investigator containing a set of questions. These questions relate to the problem of enquiry directly or indirectly. The questionnaires are mailed to the informants with a formal request to answer the question and send them back within a specified time. For better response the investigator should bear the postal charges. The questionnaire should carry a note explaining the aims and objective of the enquiry, definition of various terms and concepts used there.

Drafting of Questionnaire:

Success of this method greatly depends upon the way in which the questionnaire is drafted. So the investigator must be very careful while framing the questions. The questions should be-

(i) Short and clear.

(ii) Few in number.

(iii) Simple and intelligible.

(v) Impersonal, non-aggressive type.

(vi) Simple alternative, multiple-choice or open-end type.

(a) In the simple alternative question type, the respondent has to choose between alternatives such as 'Yes or No', 'right or wrong' etc.

4. Interview Schedule Method

In case the informants are largely uneducated and non-responsive data cannot be collected by the mailed questionnaire method. In such cases, schedule method is used to collect data. Here the interview schedules are sent through the enumerators to collect information. Enumerators are persons appointed by the investigator for the purpose. They directly meet the informants with the interview schedules. They explain the scope and objective of the enquiry to the informants and solicit their co-operation. The enumerators ask the questions to the informants and record their answers in the interview schedule and compile them. The success of this method depends on the sincerity and efficiency of the enumerators.

5. From Local Agents

Primary data are collected from local agents or correspondents. These agents are appointed by the sponsoring authorities. They are well conversant with the local conditions like language, communication, food habits, traditions etc. Being on the spot and well acquainted with the nature of the enquiry they are capable of furnishing reliable information. The accuracy of the data collected by this method depends on the honesty and sincerity of the agents. The method is generally used by government agencies, newspapers, periodicals etc.

6. Through Telephone

The researchers get information through telephone this method is quick and give accurate information.

Secondary Data

The secondary data are the second hand information which is already been collected and analysed by someone else for some purpose The secondary data are not pure in character, *for example,* Economics survey of India is secondary data because these are collected by more than one organization like Bureau of statistics, Board of Revenue, the Banks etc.

Secondary data analysis saves time that would otherwise be spent collecting data and, particularly in the case of quantitative data, provides larger and higher-quality databases that would be unfeasible for any individual researcher to collect on their own. In addition, analysts of social and economic change consider secondary data essential, since it is impossible to conduct a new survey that can adequately capture past change and/or developments.

Advantages of Secondary sources of Data

1. ***Ease of Access:*** There are many advantages to using secondary data. This includes the relative ease of access to many sources of secondary data. In the past secondary data accumulation required marketers to visit libraries or wait for

reports to be shipped by mail. Now with the availability of online access, secondary data is more openly accessed. This offers convenience and generally standardized usage methods for all sources of secondary data.

2. ***Low Cost to Acquire:*** The use of secondary data has allowed researchers access to valuable information for little or no cost to acquire. Therefore, this information is much less expensive than if the researchers had to carry out the data themselves.

3. ***Clarification of Data Question:*** The use of secondary data may help the researcher to clarify the data question. Secondary data is often used prior to primary data to help clarify the data focus.

4. ***May Answer Researcher Question:*** The use of secondary data collection is often used to help align the focus of large scale primary data. When focusing on secondary data, the researcher may realize that the exact information they were looking to uncover is already available through secondary sources. This would effectively eliminate the need and expense to carry out their own primary data.

5. ***May Show Difficulties in Conducting Primary research:*** In many cases, the originators of secondary data include details of how the information was collected. This may include information detailing the procedures used in data collection and difficulties encountered in conducting the primary data. Therefore, the detailed difficulties may persuade the researcher to decide that the potential information obtained is not worth the potential difficulties in conducting the data.

Uses of Secondary Data

i) The secondary data may be used in three ways by a researcher.

ii) Some specific information from secondary sources may be used for reference purposes.

iii) Secondary data may be used as bench marks against which the findings of a research may be tested.

iv) Secondary data may be used as the sole source of information for a research project. Such studies as Securities Market Behavior, Financial Analysis of Companies, and Trends in credit allocation in commercial banks, Sociological Studies on crimes, historical studies, and the like depend primarily on secondary data.

v) Year books, Statistical reports of government departments, reports of public organisations like Bureau of Public Enterprises, Census Reports etc. serve as major data sources for such research studies.

Sources of Secondary Data

There are mainly two sources of secondary data as follows:

1. Internal Sources of Secondary Data
2. External Sources of Secondary Data

1. Internal Sources of Secondary Data

Internal sources can be classified into four broad categories:

i) Accounting records
ii) Sales force reports
iii) Miscellaneous records
iv) Internal experts

i) Accounting Records

The basis for accounting records concerned with sales is the sales invoice. The usual sales invoice has a sizable amount of information on it, which generally includes name of customer, location of customer, items ordered, quantities ordered, quantities shipped, Rupees extensions, back orders, discounts allowed and date.

In addition, the invoice often contains information on sales territory, sales representative, and warehouse of shipment. This information, when supplemented by data on costs and industry and product classification, as well as from sales calls, provides the basis for a comprehensive analysis of sales by product, customer, industry, geographic area, sales territory and sales representative, as well as the profitability of each sales category. Unfortunately, most firms' accounting systems are designed primarily for tax reasons rather than for decision support.

ii) Sales Force Reports

Sales force reports represent a rich and largely untapped potential source of marketing information. The word potential is used because evidence indicates that sales personnel do generally not report valuable marketing information.

Sales personnel often lack the motivation and/or the means to communicate key information to marketing managers. To obtain the valuable data available from most sales forces, several elements are necessary: (a) a clear, concise statement, repeated frequently, of the types of information desired; (b) a systematic, simple process for reporting the information; (c) financial and other rewards for reporting information; and (d) concrete examples of the actual use of the data.

iii) Miscellaneous Reports

Miscellaneous reports represent the third internal data source. Previous marketing research studies, special audits and reports purchased from outside for prior problems may have relevance for current problems. As a firm becomes more diversified, the more likely it is to conduct studies that may have relevance to problems in other areas of the firm.

For example, P&G sells a variety of distinct products to identical or similar target markets. An analysis of the media habits conducted for one product could be very useful for a different product that appeals to the same target market. Again, this requires an efficient marketing information system to ensure that those who need them can find the relevant reports.

iv) Internal Experts

One of the most overlooked sources of internal secondary data is internal experts. An internal expert is anyone employed by the firm who has special knowledge. The following statement by a senior research manager at a major consumer goods firm describes why his organization developed a research reports library and how they ensure its use. On the average, each brand is assigned a new brand manager every two years.

2. External Sources of Secondary Data

Numerous sources external to the firm may have data relevant to the firm's requirements. Seven general categories of external secondary information are described in the sections that follow: i) computerized databases, ii) associations, iii) government agencies, iv) syndicated services, v) directories, vi) other published sources and vii) external experts.

i) Databases

A computerized database is a collection of numeric data and/or information that is made computer-readable form for electronic distribution. There are than 3,500 databases available from over 550 on-line service enterprises. Those that are available that are useful in bibliographic search, site location, media planning, market planning, forecasting and for many other purposes of interest to marketing researchers.

ii) Associations

Associations frequently publish or maintain detailed information on industry sales, operating characteristics, growth patterns, and the like. Furthermore, they may conduct special studies of factors relevant to their industry. These materials may be published in the form of annual reports, as part of a regular trade journal, or as special reports. In some cases, they are available only on request from the association. Most libraries maintain reference works, such as the Encyclopedia of Associations that list the various associations and provide a statement of the scope of their activities.

iii) Government Agencies

Federal, state, and local government agencies produce a massive amount of data that are of relevance to marketers. In this section, the nature of the data produced by the federal government is briefly described. However, the researcher should not overlook state and local government data.

There are also a number of specialized analytic and research agencies, numerous administrative and regulatory agencies and special committees and reports of the judicial and legislative branches of the government. These sources produce five broad types of data of interest to marketers. There are data on (1) population, housing and income; (2) agricultural, industrial and commercial product sales of manufacturers, wholesalers, retailers and service organizations; (3) financial and other characteristics of firms; (4) employment and (5) miscellaneous reports.

iv) Syndicate Services

A wide array of data on both consumer and industrial markets is collected and sold by commercial organizations.

v) Directories

Any sound marketing strategy requires an understanding of existing and potential competitors and customers. Suppose you were asked to prepare a report on the forest products industry, to aid your organization in developing a sales and marketing approach to lumber manufacturers. A number of services and directories would prove useful. A general industry directory such as Thomas Register of American Manufacturers is a good starting place. This sixteen-volume, set lists manufacturers' products and services by product category. It provides the company name, address, telephone number and an estimate of its asset size. It also contains an extensive trademark listing and samples of company catalogs.

vi) Other Published Sources

There is a virtually endless array of periodicals, books, dissertations, special reports, newspapers and the like that contain information relevant to marketing decisions.

vii) External Experts

External experts are individuals outside your organization whose job provides them with expertise on your industry or activity. State and government officials associated with the industry, trade association officials, editors and writers for trade and publications, financial analysts focusing on the industry, government and university researchers and distributors often have expert knowledge relevant to marketing problems.

Techniques of Data Collection

A statistical survey can be conducted either by the (i) census method sampling method.

1. Census Method

A census is a study that obtains data from each and every unit of a population is called census or complete enumeration. In most studies, a census is not practical, because of the cost and/or time required.

A census is the procedure of systematically acquiring and recording information about the members of a given population. It is a regularly occurring and official count of a particular population. The term is used mostly in connection with national population and housing censuses; other common censuses include agriculture, business and traffic censuses. In the latter cases the elements of the 'population' are farms, businesses and so forth, rather than people. When the whole area or population of persons is contacted the method is known as census method.

Merits of Census Method

1. Data is obtained from each and every unit of population.
2. The results obtained are likely to be more representative, accurate and reliable.
3. It can be used in various aspect of survey e.g. Indian population censes surve.
4. It can be exploited as a basis for various surveys.

Demerits of Census Method

1. The effort, money and time-extremely large.
2. Population is infinite.
3. Exhaustive and intensive study-impossible.
4. It is expensive and time consuming.

2. Sampling Method

Under this technique some representative units or informants are selected from the universe. These selected units are called samples. Based upon the data collected from these samples conclusions are drawn upon the whole universe

Principles of Sampling

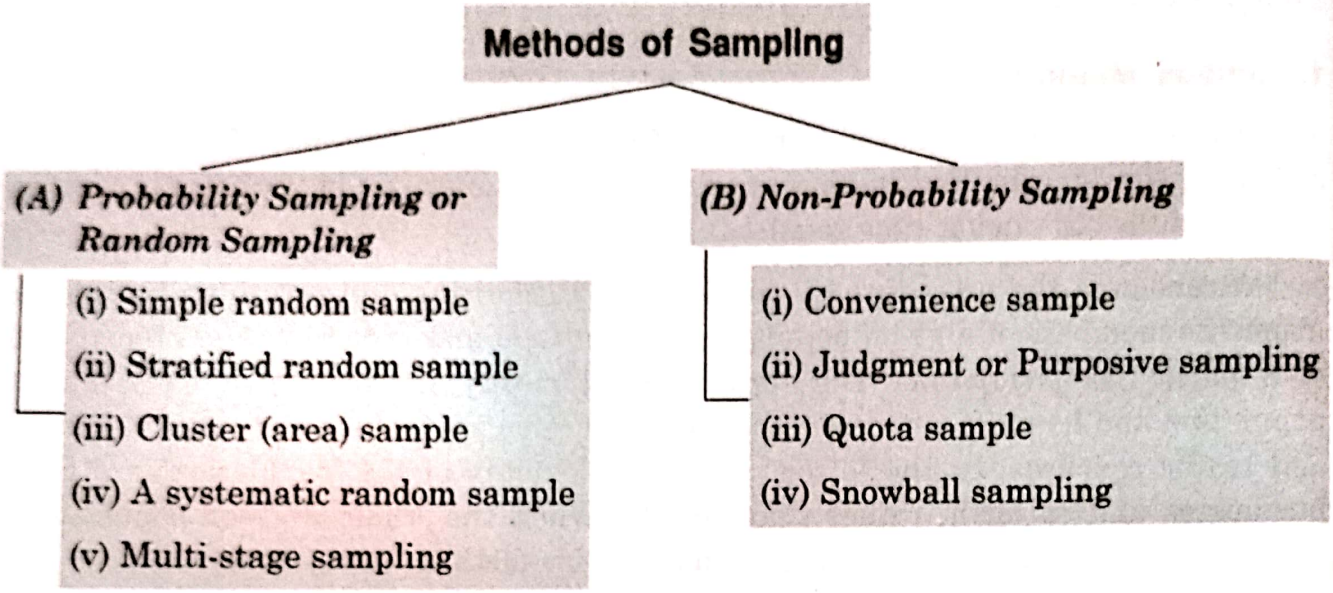
The theory of sampling is based on two important principles:

(i) ***Principal of Stastical Regularity:*** Sample is taken at random from a population it is likely to possess almost the same characteristics as that of the population.

(ii) ***Principles of Inertia of Large Numbers:*** other things being equal, large the size of the sample, more accurate the results are likely to be, i.e. Large number are more stable as compared to small ones.

Methods of Sampling

The various types of sampling can be broadly divided into two:



(A) Probability Sampling or Random Sampling

(i) Simple random sample

Every member of the population has a known and equal chance of being selected. *For example:* (i) lottery method and (ii) the use of table of Random Number.

(ii) Stratified random sample

All people in sampling frame are divided into "strata" (groups or categories). Within each stratum, a simple random sample or systematic sample is selected. *Example* if we want to ensure that a sample of 5 students from a group of 50 contains both male and female students in same proportions as in the full population (i.e. the group of 50), we first divide that population into male and female. In this case, there are 22 male students and 28 females. To work out the number of males and females in the sample No. of males in sample = (5 / 50) x 22 = 2.2 , No. of females in sample = (5 / 50) x 28 = 2.8. Therefore we choose 2 males and 3 females in the sample. These would be selected using simple random or systematic sample methods.

(iii) Cluster (area) sample

The population is divided into mutually exclusive groups such as blocks, and the investigator draws a sample of the group to interview. *For example,* a cluster may be a village or a school, a state. So you decide all the elementary schools in Karnataka State are clusters. You want 20 schools selected. You can use simple or systematic random sampling to select the schools and then every school selected becomes a cluster. If you interest is to interview teachers on the opinion of some new program which has been introduced, then all the teachers in a cluster must be interviewed.

(iv) A systematic random sample

Sample is obtained by selecting one unit on a random basis and choosing additional elementary units at evenly spaced intervals i.e., every nth item until the desired number of units is obtained. *For example,* there are 100 students in your class. You want a sample of 20 from these 100, divide 100 by 20, you will get 5. Randomly select any number between 1 and five. Suppose the number you have picked is 4, that will be your starting number from there you will select every 5th name i.e.,4th name, 9th name (4 + 5) 14th name (9 + 5) and so on until you reach the last one hundred you will end up with 20 selected students.

(v) Multi-stage sampling

As the name implies, this involves drawing several different samples. It does so in such a way that cost of final interviewing is minimised. *For example,* first draw sample of areas. Initially large areas selected then progressively smaller areas within larger area are sampled. Eventually end up with sample of households and use method of selecting individuals from these selected households.

(B) Non-Probability Sampling

(i) Convenience Sample

The investigator selects the easiest population members from which to obtain information.

(ii) Judgment Sample or Purposive Sampling

A purposive sample is one which is selected by the investigator subjectively. The investigator uses his/her judgement to select population members who are good prospects for accurate information.

(iii) Quota Sample

The investigator finds and interviews a prescribed number of people in each of several categories.

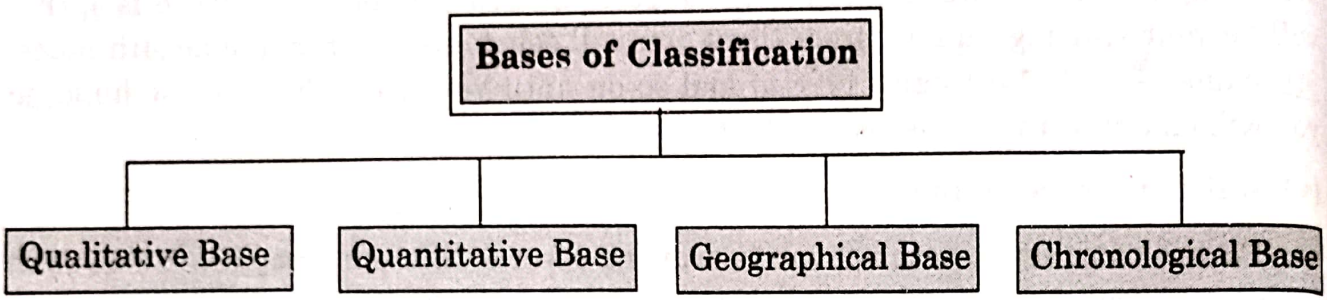
(iv) Snowball / Reference Sampling

With this approach, you initially contact a few potential respondents and then ask them whether they know of anybody with the same characteristics that you are looking for in your purpose.

For example, if you wanted to interview a sample of vegetarians / cyclists / people with a particular disability etc., you initial contacts may well have knowledge of others.

Classification

The process of dividing the data into different groups or classes which are homogeneous within but heterogeneous between themselves is called classification i.e. the process of arranging data into homogenous group or classes according to some common characteristics.



Bases of Classification or Modes of Classification

There are four important types of classification:

1. Qualitative Base
2. Quantitative Base
3. Geographical Base
4. Chronological or Temporal Base

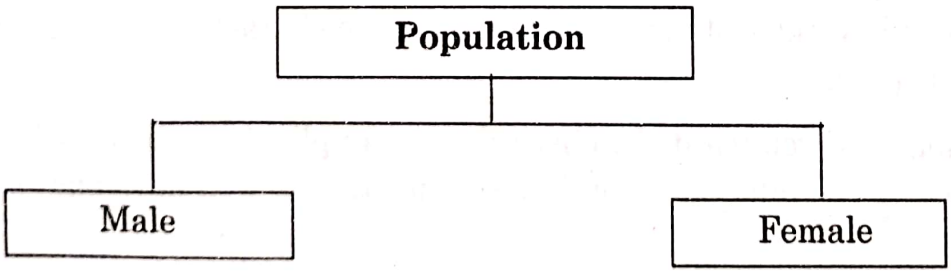
1. Qualitative Base

When the data are classified according to some quality or attributes such as sex, religion, literacy, intelligence etc. Qualitative classification may be done in:

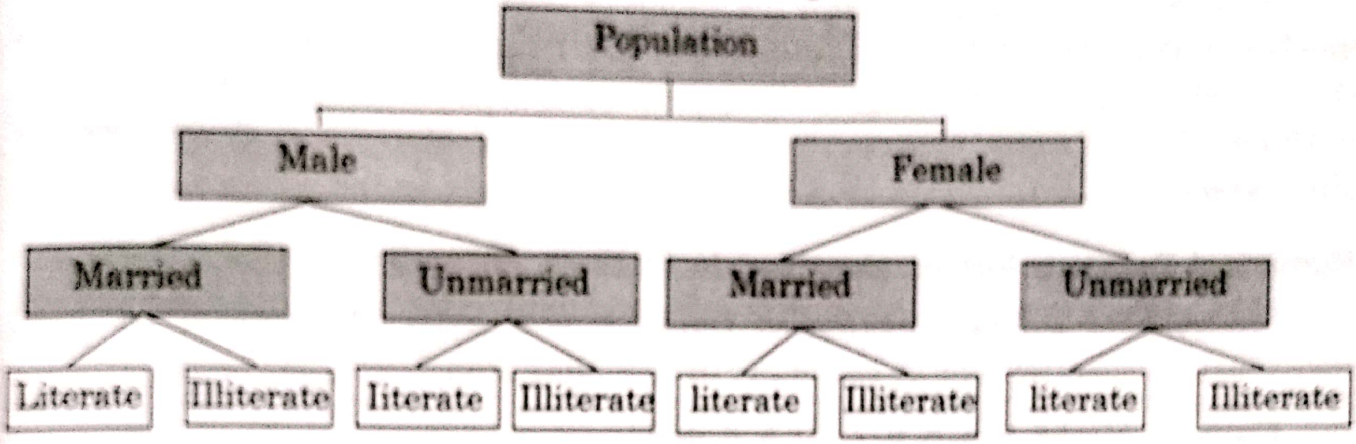
(i) Simple classification and

(ii) Manifold classification

(i) **Simple classification:** When only one attribute or Character is used for classification, is called simple classification. e.g.



(ii) **Manifold classification:** When more than one attributes or Character is used for classification is called manifold classification e.g.



2. **Quantitative Base:** When the data are classified by quantitative characteristics i.e. with measurable units like heights, weights, ages, income etc.

3. **Geographical Base:** When the data are classified by geographical regions or location, like states, provinces, cities, countries etc. *Example:*

States	*Sales (₹) in Lakhs*
Karnataka	500
Tamilnadu	450
Andrapradesh	560
Kerala	350

4. **Chronological or Temporal Base:** When the data are classified or arranged by their time of occurrence, such as years, months, weeks, days etc... *For example:*

Years	*Sales (₹) in Crores*
2012	3.58
2013	4.90
2014	5.36
2015	5.85